

Trust-Aware Bio-Inspired Swarm Defense using TA-MAPPO

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Plan of the Presentation

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Project Motivation

- ▶ Develop a trust-aware swarm defense framework for multi-drone coordination under adversarial communication.
- ▶ Build a continuous-control multi-agent reinforcement learning environment for robust navigation.
- ▶ Combine trusted physical sensing with untrusted peer communication.
- ▶ Prepare the system for future traitor-node detection and isolation.

System Overview

- ▶ Core method: Trust-Aware Multi-Agent Proximal Policy Optimization (TA-MAPPO).
- ▶ Domain: decentralized drone swarm navigation in a continuous 2D world.
- ▶ Key challenge: maintain safe coordination despite noisy or malicious neighbor broadcasts.
- ▶ Strategy: centralized training, decentralized execution.

Architectural Foundation

- ▶ Built on the PettingZoo ParallelEnv API.
- ▶ A custom VecEnv wrapper converts multi-agent dictionary outputs into batched arrays for Stable-Baselines3.
- ▶ Environment space: continuous 20×20 bounded 2D region.
- ▶ Action space: continuous thrust vector (v_x, v_y) with range $[-1, 1]$.
- ▶ Time integration step: $\Delta t = 0.1$ seconds, with an 800-step episode cap.

CTDE Framework

- ▶ Training paradigm: Centralized Training with Decentralized Execution (CTDE).
- ▶ Actor network operates only on local observations.
- ▶ Critic network operates on full global state information.
- ▶ Actor input dimension: 130.
- ▶ Critic input dimension: 520.
- ▶ Both actor and critic use custom MLP feature extractors.

Core Hyperparameters

- ▶ RL algorithm: Proximal Policy Optimization (PPO).
- ▶ Learning rate: 5×10^{-5} with linear decay.
- ▶ Entropy coefficient: 0.005 in early training, 0.03 in exploratory hardening phases.
- ▶ Parallel rollout: 10 CPU workers handling 10 drones each.
- ▶ Reward normalization enabled; observation normalization disabled to preserve geometry.

Dual-Sensing Paradigm

- ▶ Trusted sensing: LiDAR raycasting provides direct geometric boundary awareness.
- ▶ Untrusted sensing: neighbors broadcast state information such as position and velocity.
- ▶ Honest drones may be misled by false communication from traitor drones.
- ▶ Future trust module will estimate pairwise trust coefficients T_{ij} .

Trust-Gated Communication

Incoming communication is filtered before policy use:

$$h'_j = h_j \odot T_{ij}$$

- ▶ h_j denotes received communication features from neighbor j .
- ▶ $T_{ij} \in [0, 1]$ represents trust assigned by drone i to drone j .
- ▶ Trusted physical sensing bypasses this gate.
- ▶ This creates resilience against topological deception attacks.

Conclusion

- ▶ The environment improved from an early 21% success regime to near-99% robustness.
- ▶ The system now supports optimized large-scale multi-agent continuous-control training.
- ▶ Phase C and D will complete the trust-aware adversarial defense layer.
- ▶ TA-MAPPO provides a strong foundation for resilient swarm autonomy.

Thank You